YUHAN QIAO

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EDUCATION

Beijing JiaoTong University, Beijing, China *Sep 2020 - Jun 2024*
M.Eng. in Computer Science & Technology
Average Score: **87.6/100**, GPA: **3.65/4.00**

China University of Mining & Technology-Beijing, Beijing, China *Sep 2016 - Jun 2020*
B.Eng. in Information Engineering
Average Score: **90.5/100**, GPA: **3.82/4.00**

LANGUAGE PROFICIENCY

- English: **TOEFL iBT: 106/120** (R:**30/30**, L:26, S:23, W:27) *Test Date: Jun 2025*
- Mandarin: Native

RESEARCH INTERESTS

- Trustworthy and robust AI
- Graph representation learning and social media analysis
- Semi-/self-supervised learning for low-resource settings
- Rumor and misinformation detection on social media

PUBLICATION

- A Debiased Self-Training Framework with Graph Self-Supervised Pre-training Aided for Semi-Supervised Rumor Detection
  Yuhan Qiao, Chaoqun Cui, Yiying Wang, Caiyan Jia
  Neurocomputing, Aug 2024. (SCI, IF = 6.5, JCR Q1).
  doi:10.1016/j.neucom.2024.128314
- Rumor Detection on Social Media based on Graph Contrastive Self-Supervised Learning (in Chinese)
  Yuhan Qiao, Caiyan Jia
  Journal of Nanjing University (Natural Sciences), Sep 2023.
  (Originally presented orally at the 19th ***China Conference on Machine Learning (CCML 2023)***, later recommended for journal publication)
  doi:10.13232/j.cnki.jnju.2023.05.010

RESEARCH EXPERIENCE

Master's Thesis | Advisor: Prof. Caiyan Jia *Dec 2020 - Jun 2024*
Research on Rumor Detection Methods based on Graph Representation Learning
Beijing Key Lab of Traffic Data Analysis and Mining, Beijing Jiaotong University Beijing, China

- **Summary:** This research investigated rumor detection through the lens of graph representation learning. Rumor propagation structures were modeled as graphs, and self-supervised and semi-supervised frameworks were developed to improve the robustness and generalization of rumor detection models, addressing overfitting issues caused by limited labeled data in real-world rumor scenarios.
- **Project 1:** Graph Contrastive Self-Supervised Learning Framework for Rumor Detection
  - Modeled rumor propagation structures as graphs and proposed a contrastive self-supervised learning framework to enhance robustness and generalization of GNN-based rumor detection models.

  - Designed graph-level augmentation strategies based on rumor propagation patterns and integrated the contrastive and supervised classification tasks into a single end-to-end framework.
  - Achieved consistent performance gains across three public benchmarks, demonstrating that the proposed contrastive framework can serve as a general add-on to enhance model robustness and generalization.
  - Presented this work orally at *CCML 2023* and subsequently published it in the *Journal of Nanjing University (Natural Sciences)*.

- **Project 2:** Debiased Self-Training for Semi-Supervised Rumor Detection
  - Proposed a self-training framework for rumor detection to reduce dependence on labeled rumor data, and designed several debiasing strategies to reduce noise accumulation during the self-training process.
  - Leveraged both contrastive and generative graph self-supervised methods to enhance the initial model used for self-training, thereby reducing erroneous pseudo-labels in the early stages of self-training.
  - Designed a pseudo-labeling strategy inspired by curriculum learning, featuring self-adaptive global and class-specific local thresholds to improve pseudo-label quality during the iterative self-training process.
  - Conducted extensive experiments on four public benchmarks under semi-supervised settings, showing consistently superior performance. Notably, we achieved 86.0% accuracy on Weibo with just 5 labels per class.
  - Published this work in *Neurocomputing* and identified promising directions, including handling class imbalance and developing consistency-regularization-based approaches for semi-supervised rumor detection.

Research Intern | Advisor: Prof. Zhineng Chen *Oct 2019 - Nov 2020*
Introductory Machine Learning & Computer Vision Project Series
Institute of Automation, Chinese Academy of Sciences Beijing, China

- **Summary:** Received foundational training for later research by implementing classical computer vision and deep-learning pipelines across several small-scale projects.
- **License Plate Recognition:** Implemented a license-plate recognition mini project using OpenCV and a KNN classifier, including image preprocessing, character segmentation and classification.
- **Face Detection and Recognition Series:**
  - Developed face-detection systems using both skin-color models and AdaBoost-based cascade classifiers with Haar-like features.
  - Applied the Eigenfaces (PCA) method for face recognition on the Yale2 dataset.
  - Implemented neural network–based classifiers, including fully connected networks on MNIST and CNN models such as LeNet-5 and AlexNet on the Caltech101 dataset.

AWARDS AND HONORS

- 2020–2022: 1st-Class Academic Scholarship, Beijing JiaoTong University (BJTU)
- 2019: Outstanding Participant, BJTU National Summer Camp (led to Direct M.Eng. Admission)
- 2016–2019: Outstanding Student Scholarship, China University of Mining & Technology-Beijing
  - 2018–2019: 3rd-Class
  - 2016–2018: 2nd-Class
- 2017: Second Prize, ”FLTRP CUP” English Reading Contest (Preliminary Round)

SKILLS

- **Programming Languages:** Python, C/C++ (basic)
- **Machine Learning & Tools:** PyTorch, PyTorch Geometric, DGL, OpenCV, NumPy, Scikit-learn, Matplotlib, Linux (command-line), Jupyter Notebook, LaTeX, Markdown